

ESI.ts

The History & Evolution of a TypeScript EVE Online ESI Client

`@lgriffin/esi.ts` · created **19 June 2024** · 58 commits on `main` · 243
merged PRs · v10.2.1

Repository at a Glance

Metric	Value
First commit	2024-06-19 ("Initial Structure")
Commits on main	58
Authors	1 (Leigh Griffin)
Merged PRs	243 (of 285 total)
Open PRs / closed unmerged	1 / 41
Open issues	50
Local branches	189 (94 remote)

The Timeline

2024-06 repo created – auth flow, client builder, 1st stable release
2024-07 rapid API build-out: characters, contracts, mail, sov, fleets
market, killmails, industry – v1.0.0 (Aug 2)
2024-08 handler redesign, small refactors
2024-09 pagination added, npm rename (@lgriffin/esi.ts)
2024-10 docs integrated (PR #1)
— quiet period —
2025-09 major refactor (PR #2), ETag caching, CI overhaul
2025-10 universe route added
— quiet period —
2026-03 → 2026-09 the industrial era: release-please, mutation
testing, contract fuzzing, BDD overhaul, v4 → v10

Phase 1 — Foundation (Jun–Aug 2024)

- **Jun 19:** "Initial Structure" — repo born
- **Jun 27:** first stable release with a working **OAuth auth flow**
- **Jun–Jul:** client-builder refactor; 20+ ESI domains added in ~4 weeks
(alliances, characters, corporations, contracts, mail, skills, PI, sov, fleets, fittings, killmails, industry, market, dogma, route, status...)
- **Aug 2: v1.0.0 complete**
- Style: direct commits, one author, tests fixed as they went

Phase 2 — Hardening (Sep 2024 – Oct 2025)

- **Pagination** added (Sep 2024) + **npm rename** to `@lgriffin/esl.ts`
- **Docs integrated** into the build (Oct 2024, PR #1)
- **Sep 2025:** the "Major Refactor" (PR #2, *director* branch):
 - **ETag / 304 caching** for bandwidth savings
 - CI/CD modernization, version-bump automation
- **Oct 2025:** universe route — last commit of the era

Phase 3 — The Industrial Era (2026)

A change of scale: PRs per month explode.

Month	PRs
2026-03	5
2026-04	14
2026-05	5
2026-06	46
2026-07	38
2026-08	78
2026-09	65

The Numbers That Matter

243	merged PRs	189	local branches
58	commits on main	50	open issues
20	release tags	575	test cases (it/test)
213	endpoint files	290	test files
1	author	370	exported symbols

Architecture

Layered, DI-based clean architecture under `src/`:

```
src/  
├─ api/           213 endpoint modules in 32 domain folders  
│                 (alliances, characters, market, universe, ...)  
├─ clients/       32 facade clients (AllianceClient, MarketClient, ...)  
├─ core/  
│   ├─ ApiClient / ApiClientBuilder / EsiClientBuilder  
│   ├─ ApiRequestHandler → axios transport, status mapping (420, 429, ...)  
│   ├─ cache/ETagCacheManager → 304 responses, bandwidth savings  
│   ├─ container/Container → dependency injection  
│   ├─ errors/ApiError → typed error surface (./errors)  
│   └─ logger/ winston → Loggly bulk transport  
├─ builders/      client composition (EsiApiFactory, CustomEsiClient)  
├─ testing/       TestDataFactory, TestScenarios, TestAssertions (exported!)  
└─ types/         APT response contracts
```

Error Handling

- `ApiError` hierarchy with **typed error categories** (`NOT_FOUND_ERROR`, `NETWORK_ERROR`, ...) and an explicit `./errors` package surface — pinned by **type-level mutation tests** so the public error contract can't silently change
- HTTP status → message mapping table (201/204/304/400/401/403/404/420/422/429/5xx)
- **Retry logic** with injectable strategy, **circuit breaker**, and per-group rate limiting — 5 defects in this layer were *found by model-based property tests* and fixed (PR #340)

Testing Strategy — a 9-Tier System

1. **TDD / unit** — 575 test cases, 290 test files, jest + ts-jest
2. **BDD** — Given/When/Then scenario tests; every scenario must execute,
JUnit cases named by Rule, `@bug` links to tracker (PR #333)
3. **Improved / resilient** — retry, concurrency, error-recovery scenarios
4. **Contract** — recorded live ESI payloads replayed through the client pipeline; shape changes fail loudly (PRs #334, #370, #386)
5. **Property / model-based** — caught 5 real defects (circuit breaker, pagination cache, backoff)

CI/CD — 20 Workflows

PR gate:	CI Fast · CodeQL · zizmor (local) · mutation (changed files only)
Nightly:	Mutation Testing · Mutation Retry · Fault Tier · Property Run Schemathesis Fuzz · ESI Spec Drift · Recorded Payloads Interleaving · No-Retry Run · Benchmarks & Heap Soak · Security Audit
Release:	Release Please · Release Pipeline · SBOM (signed CycloneDX) Post-publish Canary · "verify what npm serves"
Quality:	OSSF Scorecard · Dependabot · Maintenance & Security

Latest 30 runs: **25 success · 2 failure · 1 skipped · 2 action-required.**

Release & Security Posture

- **20 tags, v4.0.0 → v10.2.1** — release-please driven, semver rules pinned against drift
- **Signed SBOM** (CycloneDX) attached to every release (PR #372)
- **Publish-the-tested-tarball**: verify the artifact npm actually serves (PRs #364, #365)
- **Post-publish canary** validates a real consumer install
- **zizmor** workflow audit + OSSF Scorecard improvements + workflow hardening
- Pinned dependencies, lockfile sync, provenance checks

Evolution in One Slide

2024	"make it work"	→ 20+ ESI domains in 4 weeks, v1.0.0
2025	"make it solid"	→ major refactor, ETag caching, CI
2026	"make it provable"	→ 9-tier testing, mutation floors, contract replay, signed SBOM, v10.x

The same 307 source files. A very different guarantee.

Questions?

ESI.ts — `npm install @lgriffin/esi.ts`

58 commits, 243 PRs, 9 test tiers, 1 obsessive author.